American Journal of Epidemiology
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Vol. 161, No. 5
Printed in U.S.A.
DOI: 10.1093/aje/kwi059

ORIGINAL CONTRIBUTIONS

Secondhand Smoke, Dietary Fruit Intake, Road Traffic Exposures, and the Prevalence of Asthma: A Cross-Sectional Study in Young Children

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Received for publication August 23, 2004; accepted for publication November 19, 2004.

The authors have investigated the independent effects of exposure to secondhand smoke, road vehicle traffic, and dietary fruit intake in a cross-sectional study of asthma in young children. They surveyed all children aged 4–6 years in 235 schools in the East Midlands and East of England regions of the United Kingdom in 2003. Data on respiratory symptoms, diagnoses and treatment, smoking in the home, and dietary fruit intake were collected by parental questionnaire. A geographic information system was used to map postcodes and determine the distance of the home from the nearest main road. Responses were obtained from 11,562 children. Wheeze in the past year and physician-diagnosed asthma were reported by 14.1% and 18.2%, respectively. Both of these outcomes were more common in children who lived with a smoker, and the prevalence of asthma increased with the number of smokers in the home. Asthma prevalence was not associated with proximity of the home to a main road or with dietary fruit intake. The authors conclude that, of the potential risk factors considered in this study, preventing secondhand smoke exposure may be the most effective way of preventing asthma.

asthma; diet; public health; tobacco smoke pollution; vehicle emissions

Exposures to secondhand tobacco smoke, road vehicle traffic, and diet are some of the most prevalent modifiable risk factors for asthma in children. The effect of parental smoking on wheezing illness and diagnosed asthma in children is well established (1, 2), but evidence that these outcomes are more common in children living close to a main road (3–5) has not been confirmed in all studies (6, 7). Several dietary factors have been linked to asthma (8), and one of the most consistent observations is of an inverse association with fruit intake (9–13).

The National Schools Fruit Scheme is a government initiative that aims to provide each child aged 4–6 years with free fruit in school every day by winter 2004. As part of an evaluation of the health benefits of this scheme, we have taken the opportunity to investigate the relative importance of fruit intake, exposure to secondhand smoke, and road vehicle traffic in determining the prevalence of asthma in over 11,000 children.

MATERIALS AND METHODS

Participants comprised all eligible children attending a sample of schools in two regions of England, the East Midlands and Eastern Region, in the summer 2003 term, before commencement of the National School Fruit Scheme. We contacted a random sample of 225 schools in each region, with the aim of recruiting 125 in each region, and distributed a short questionnaire to parents of all children aged 4–6 years. We included questions relating to wheeze and eczema, based on standard wording from the International Study of Asthma and Allergies in Childhood (14). We also asked about asthma diagnosed by a physician and whether the child had a reliever or steroid inhaler for asthma and, if so, how many puffs he or she had used in the past week. The questionnaires were scanned and entered into a database (Document Capture Company, Wembley, United

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TABLE 1. Distribution of asthma outcomes by age, sex, local education authority area, Townsend Index, and presence of eczema, United Kingdom, 2003

	Frequency (no.)	%	Wheeze in past year (%)*	Physician-diagnosed asthma (%)†	Asthma medication (%)‡
Sex					
Male	5,835	50.5	16.5	21.1	18.7
Female	5,711	49.5	11.6	15.2	13.2
p value			<0.001	<0.001	<0.001
Age (years)					
4	1,625	14.1	14.9	16.6	16.1
5	5,780	50.1	14.4	17.5	15.5
6	4,130	35.9	13.4	19.7	16.5
p_{trend}			0.1	0.002	0.4
Area					
East Midlands	5,567	48.1	13.8	17.8	15.6
Eastern Region	5,995	51.9	14.3	18.5	16.3
p value			0.5	0.3	0.3
Townsend Index					
1 (least deprived)	2,091	20	11.7	16.7	15.1
2 (–3.4637 to –2.1311)	2,083	20	14.1	17.8	16.2
3 (–2.1310 to –0.4519)	2,088	20	14.1	17.4	15.4
4 (–0.4518 to 1.7336)	2,088	20	14.0	18.3	16.1
5 (most deprived)	2,087	20	16.9	20.9	17.5
p_{trend}			<0.001	0.001	0.07
Eczema in past year§					
No	9,140	80.5	11.5	15.7	13.4
Yes	2,216	19.5	24.9	28.2	26.5
p value			<0.001	<0.001	<0.001

* Wheeze in the past year defined as a positive response to the question: "Has your child had wheezing or whistling in the chest in the last 12 months?"

† Diagnosed asthma defined as a positive response to the questions: "Has your child ever had asthma?" and "Has this been confirmed by a physician?"

‡ Asthma medication defined as a positive response to one of the following questions: "Does your child have a reliever inhaler?" or "Does your child have a preventer inhaler?" (Both were given with examples.)

§ Eczema in the past year defined as a positive response to both of the following questions: "Has your child ever had an itchy skin rash that has affected the skin creases (e.g., folds of elbows, behind knees) at some stage?" and "Has your child had this rash in the past 12 months?"

Kingdom). Ethics permission was obtained from the Eastern Multiple Regional Ethics Committee.

Exposure to secondhand smoke was elicited by asking how many people living in the child's household smoke, and exposure to fruit intake by asking how many days in a typical week the child eats fruit, excluding fruit juice, and how many pieces of fruit on average per day. These were multiplied together to estimate the number of portions of fruit per week, categorized for analysis as no fruit intake, less than 7, 7–13, 14–20, or 21 or more. The number of smokers in the household was categorized as none, 1, 2, or 3 or more. We asked specifically about the number of apples eaten per week. Parents provided their postcode, which was linked using the postcode-enumeration district directory (Manchester Information and Associated Services, University of Manchester, Manchester, United Kingdom) to census enumeration district, and in turn to the Townsend *Z* score for the 1991 census, as a marker of deprivation. The Townsend Index was analyzed in quintiles. The postcode was also used to estimate the distance of the child's home to the nearest main road, by converting the code to northing and easting grid references of 1-m resolution, using "code point" software (Ordnance Survey, Southampton, United Kingdom) in the EDINA national data center (The University of Edinburgh, Edinburgh, Scotland). We then linked this grid reference to a digitized map of Great Britain with a coordinate resolution of 1 m (Strategi database; Ordnance Survey), including all main roads, defined as motorways, A- or B-class roads,

TABLE 2. The univariate and multivariate effects of smoking in the home, fruit intake, and distance from main road upon wheeze in the past year, United Kingdom, 2003

	Frequency (no.)	%	Wheeze in past year (%)	Age, sex, and area adjusted		Fully adjusted*	
				Odds ratio	95% confidence interval	Odds ratio	95% confidence interval
No. of smokers in the home							
0	7,429	65.2	13.0	1		1	
1	2,606	22.9	15.2	1.21	1.07, 1.37	1.18	1.02, 1.35
2	1,188	10.4	18.6	1.54	1.31, 1.81	1.40	1.17, 1.69
≥3	164	1.4	18.3	1.52	1.04, 2.23	1.47	0.97, 2.23
p_{trend}			<0.001				
Distance of home from main road (m)							
≥150	8,461	73.2	13.8	1		1	
90–149	1,297	11.2	15.7	1.14	0.97, 1.35	1.14	0.96, 1.36
30–89	1,212	10.5	14.5	1.05	0.89, 1.23	1.02	0.87, 1.21
<30	592	5.1	13.2	0.95	0.73, 1.23	0.90	0.69, 1.18
p_{trend}			0.6				
Dietary fruit intake (portions per week)							
≥21	1,804	16.0	15.9	1		1	
14–20	2,962	26.3	14.1	0.86	0.73, 1.02	0.84	0.70, 1.01
7–13	2,625	23.3	12.8	0.77	0.65, 0.92	0.77	0.64, 0.93
1–6	3,349	29.8	13.6	0.82	0.69, 0.96	0.80	0.67, 0.95
0	507	4.5	17.0	1.02	0.79, 1.31	1.01	0.77, 1.33
p_{trend}			0.2				

*Adjusted for all the factors in the table plus the Townsend score and area.

using ArcView 3.3 geographic information system software (Environmental Systems Research Institute, Inc., Redlands, California). Individuals living within 150 m, and then in turn within 30, 60, 90, and 120 m, of a main road were identified, and categories were chosen to reflect the relation between primary pollutant exposures and distance from the road (3).

Analysis was by chi-squared tests and tests for trend, as well as by multiple logistic regression in STATA statistical software (Stata Corporation, College Station, Texas), with standard errors adjusted for cluster sampling by school. A p value of less than 0.05 (two-sided test) was considered statistically significant. We adjusted for age, sex, and area (East Midlands or East of England) as a priori confounders and then looked at the independent effects of number of smokers in the household, distance from roads, and portions of fruit in a mutually adjusted model, with additional adjustment for the Townsend Index.

RESULTS

Of the total of 450 schools (225 in each region) contacted to take part in the study, 215 (48 percent) did not respond to or refused our invitation; this was largely due to the very short time interval between award of funding for the study and the start of the fruit scheme in some schools. We therefore recruited a total of 235 schools, 113 and 122 schools in the East Midlands and Eastern Region, respectively. From an estimated 21,289 eligible children in these schools, responses were obtained from 11,562 (54 percent). Demographic data are shown in table 1; study participants were slightly more affluent than the national average (median Townsend Z score: –1.33; interquartile range: –3.16 to 1.10).

Wheeze in the past year was reported in 14.1 percent of children, ever wheezing in 25.3 percent, and a physician diagnosis of asthma in 18.2 percent; 16 percent of children were reported to have an asthma medication (15 percent a bronchodilator and 10 percent inhaled steroids), and 7.9 percent had used asthma medication in the last week. All of these outcomes were more common in boys (table 1) and in relation to greater deprivation. A total of 19 percent of children had had eczema symptoms in the past year.

Thirty-five percent of children lived in a household with at least one smoker. In univariate analysis, living with a smoker was significantly associated with the child's having wheezed in the past year and with a physician diagnosis of asthma (data not shown), and both of these outcomes were more common with increasing numbers of smokers in the home (both $p < 0.001$) (table 2). Twenty-seven percent of children lived within 150 m of a main road and 5 percent within 30 m. There was no evidence of any association between distance from a main road, analyzed as either a factor or a trend, and either wheeze in the past year or diagnosed asthma (table 2). The median number of portions of fruit eaten per week was 10 (interquartile range: 5–14 portions); 49 percent of children ate fruit less than once a day, and 4 percent never ate fruit. Wheeze in the past year was most common in those

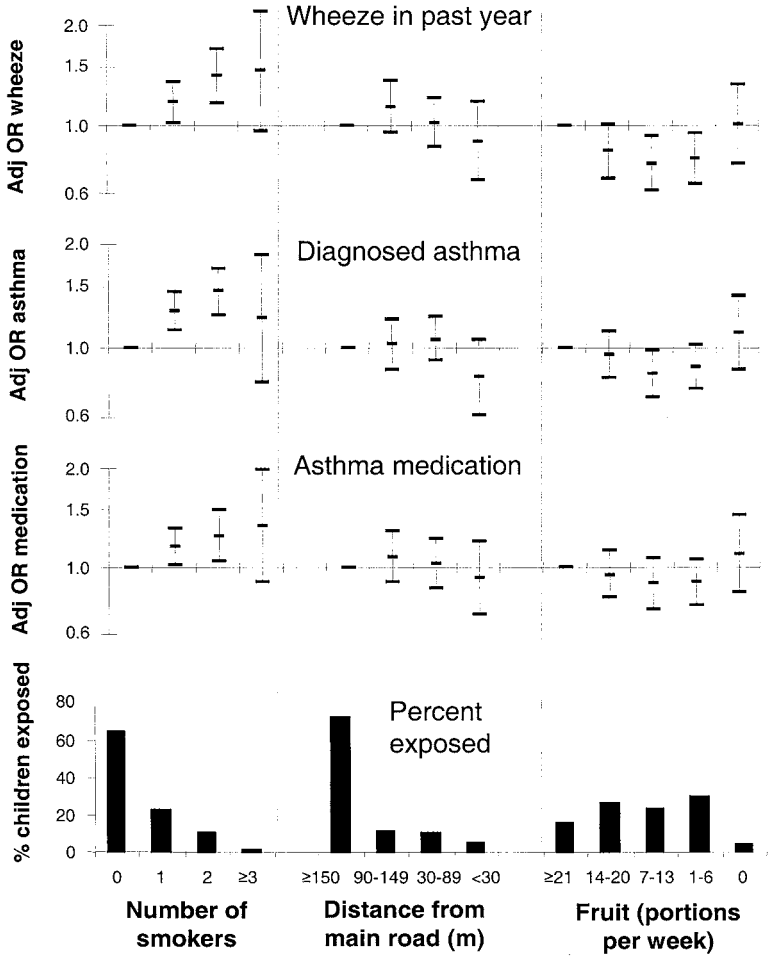


FIGURE 1. Fully adjusted odds ratios for the effect of the number of smokers in the household, distance from the main road, and number of portions of fruit per week on wheeze in the past year, diagnosed asthma, and having an asthma medication among 11,562 children, United Kingdom, 2003. Odds ratios are adjusted for all of the variables shown, plus age, sex, area (East Midlands or East of England), and Townsend score, and are plotted on a log scale, with 95% confidence intervals. Adj, adjusted; OR, odds ratio.

who seldom or never ate fruit, but it was also more common in those in the highest category of fruit intake (heterogeneity across categories of fruit intake: $p = 0.016$), and there was no significant trend in risk across the categories ($p = 0.24$) (table 2). A similar pattern was seen for physician-diagnosed asthma. Similar findings arose from analysis of data specific to apple intake; wheeze in the past year occurred in 16.5 percent of children who never ate apples and in 13.3 percent and 15.5 percent of those who ate 1–4 and five or more apples per week, respectively ($p_{trend} = 0.4$).

Adjusting for age and sex, area, or Townsend score made no substantial difference to these findings. In a multivariate logistic regression model including number of smokers in the home, distance from a main road, and either fruit or apple intake, only the number of smokers in the home had an independent effect on wheeze in the past year ($p_{trend} < 0.001$) (table 2). There were no significant interactions between the effects of these variables. The number of smokers in the home was also the only independent predictor of physician-diagnosed asthma and prescription of asthma medication (figure 1). Eczema was significantly inversely related to the number of smokers in the home and unrelated to distance from the road or to fruit intake (data not shown).

DISCUSSION

Our data confirm that over one in six young children in the United Kingdom currently has diagnosed asthma. Although asthma in this age group is likely to include more than one disease phenotype (15), most children with wheezing symptoms in our study had been given a diagnostic label of asthma and prescribed an asthma medication. Therefore, when we used these as alternative measures for asthma in this study to overcome the potential for misclassification or bias arising from any one measure, our findings were similar for all of these outcomes. Our findings are therefore pragmatically valid in relation to the clinical asthma phenotype. The consistent relation between all of these outcomes and

smoking in the home, as well as the demonstration of an exposure-response relation between smoking and asthma, indicates that this association is likely to be causal. If so, and given the high prevalence of exposure (35 percent), secondhand smoke exposure emerges as the most important of these three risk factors for asthma in young children. In contrast, our study found little effect of distance of the home from a main road or of dietary fruit intake.

Just over 50 percent of children in schools that agreed to participate in the study returned completed questionnaires suitable for analysis. Although the prevalence of wheezing (16, 17) and the estimated fruit intake (18) in our study population were comparable with other data from the United Kingdom for young children, the proportion of children exposed to smoking at home was slightly lower than the 42 percent reported nationally (19), possibly reflecting a poorer response from smokers. However, while the generally low response and potential bias within it may have influenced our prevalence estimates, they are relatively unlikely to have affected estimates of the associations with asthma found in our data.

The effect estimates for all of our exposures may, however, have been influenced by reporting bias. A tendency for parents of asthmatic children to overreport their child's fruit intake because of increased health awareness might have led to a reduction in the apparent protective effect of fruit. By the same reasoning, the effect of smoking could have been underestimated by systematic underreporting of smoking by parents aware of the potential effects of smoking on asthma in their child. It should also be appreciated that, since the prevalence of all of our asthma outcome measures was rather more than 10 percent, the odds ratios may tend to overestimate the true prevalence ratios.

That parental smoking is detrimental to children's respiratory health is well established, and recent meta-analyses have demonstrated that parental smoking increases the risk of respiratory illness in early infancy by a ratio of 1.5 (1, 2) and of asthma in school-age children by a ratio of 1.2 (1). Though maternal smoking may have a greater impact, paternal smoking also has an independent effect (1). Our data provide further confirmation that the prevalence of asthma in children increases with the number of smokers in the home. We estimate that, assuming a causal relation, 8 percent of asthma in children of this age is attributable to secondhand smoke exposure at home.

We found no association between living close to a main road and the prevalence of asthma. The postcodes used to georeference the child's home apply to the midpoint of about 15 houses, and this will have led to some nonsystematic misclassification of the distance from the road and, thus, to an underestimate of the magnitude of any effect. In common with most other studies, we used an objective but proxy measure for exposure to exhaust emissions based on distance from the main road alone, rather than modeled estimates of ambient air pollution. Our estimate may be a poor indicator of individual exposure to specific pollutants such as diesel or ozone, which have been implicated in asthma (20) and which will additionally depend on the volume, flow, and type of traffic on the road; pattern of air dispersion of individual pollutants; and the amount of time spent at home and exposure elsewhere. The majority of previous studies in children, using various objective measures of exposure to vehicle traffic at home, report positive associations with the prevalence of asthma or respiratory symptoms (3, 4), but these effects have not always reached conventional statistical significance (5) and have sometimes been confined to girls (5). Several other studies have found no effect (6, 7). Our findings add further evidence that road traffic exposure does not have a major influence on asthma risk in young children.

Dietary fruit intake has been related to increased lung function and reduced risk of wheeze and asthma in adults (11, 12); apples in particular seem to be important (21, 22). Findings in children are generally less consistent (9, 10, 13, 22, 23). Our data suggest that the minority of children who consume no fruit, or specifically no apples, may be at a moderately increased risk of asthma, but we found no evidence of any reduction in prevalence with higher levels of fruit intake. Using a parental questionnaire to ascertain diet is an imperfect measure, but the pattern of association with the number of apples eaten per week, which is easier to measure, was remarkably similar to that for any fruit. That our study was cross-sectional is a further important limitation, and it will be important to follow up these children after the introduction of fruit in school to assess whether altering dietary consumption of fruit has any long-term influence on respiratory health.

In conclusion, we have studied three of the potentially avoidable causes of asthma in children and found that, of these, secondhand smoke exposure was the single independent determinant of disease. On this evidence, we suggest that of the three factors considered in this study the most effective means of preventing asthma in children is likely to be to reduce exposure to secondhand tobacco smoke at home.

ACKNOWLEDGMENTS

This study was funded by the Department of Health of the United Kingdom.

The authors thank the school-teaching and secretarial staffs who made the survey work possible. Grid reference data and digitized road data are held by copyright of the Crown and were made available by the Ordnance Survey, via the code point software and Strategi database. Access to the data relating to the social deprivation indices was provided by Manchester Information and Associated Services.

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